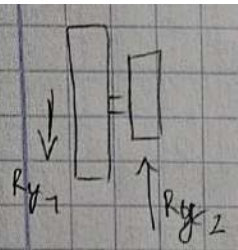
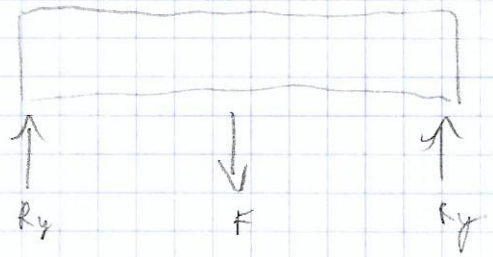


$$R_{y_2} = \left[\frac{500\,kg + 78{,}5\,kg}{4} + 0{,}18\,m \cdot 0{,}78\,m \cdot 0{,}01\,m \cdot \rho_{staal}\right] g$$

$$= 1526{,}9\,N$$ → alle opties van winkel. de mogelijk

I - ligger

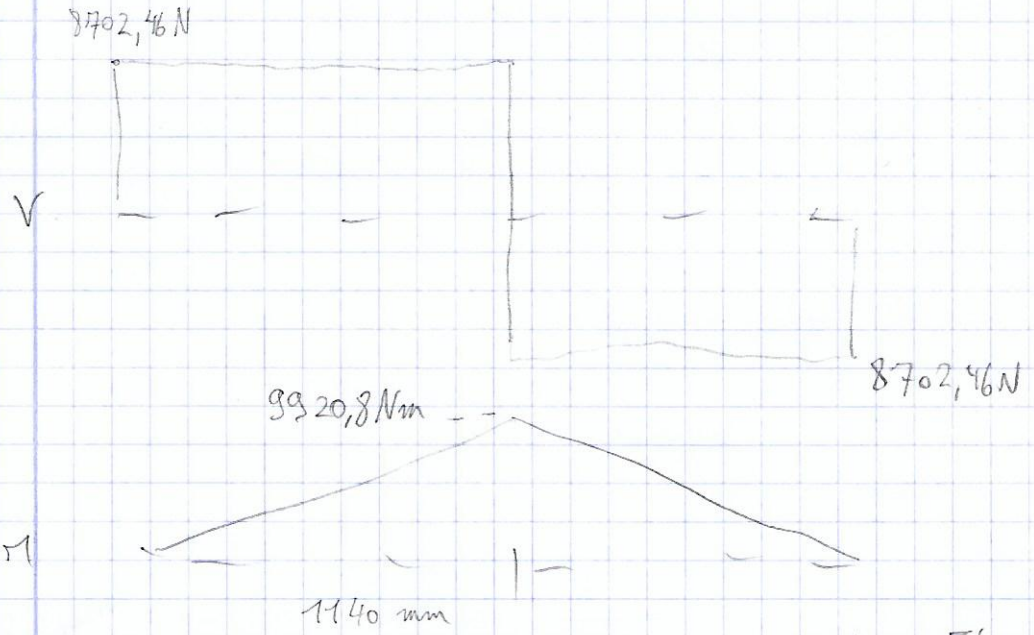


$$F = g\left[\frac{2 \cdot m_{-\,geleiding}}{3} + m_{-\,L\text{-stuk}} + m_{-\,bevestiging\text{-}ruben} + m_{-\,tandlat_Ruben} + m_{-\,AGV} + \frac{m_{-\,al\ de\ rest}}{3}\right]$$

$$= g\left[\frac{2 \cdot 32{,}1 \cdot 12}{3} + 0{,}07028 \cdot 0{,}948 \cdot 1975 \cdot 7850 + 38 + 230 + 7850 \cdot 1 \cdot 1 \cdot 0{,}05 + 4 \cdot 7850 \cdot 0{,}18 \cdot 0{,}78 \cdot 0{,}03 + 4 \cdot 5{,}45 + \frac{1}{3}\left(16 \cdot m_{\text{lochprofiel}} + 4 \cdot m_{\text{vierkant}} + 2 \cdot m_{\text{dwarsbalk}} + 4 \cdot m_{\text{rechthoek}}\right)\right]$$

$m_{\text{lochprofiel}} = 0{,}000665 \cdot 7850$; $m_{\text{vierkant}} = 7850 \cdot 0{,}4 \cdot 0{,}4 \cdot 0{,}015$; $m_{\text{dwarsbalk}} = 7850 \cdot 0{,}15 \cdot 0{,}172\,2865 \cdot 0{,}813$; $m_{\text{rechthoek}} = 7850 \cdot 0{,}3 \cdot 0{,}45 \cdot 0{,}015$

$$F = g\,(1774{,}2\ kg) = 17\,404{,}93\ N$$



$$W_{min} = \frac{F \cdot (2440 \cdot 10^{-3})}{4} \cdot \frac{1{,}5}{235\,MPa} = 59\,436 = 0{,}000\,059\,436\ m^3$$

eurocode applied

1cm

$$\delta = \frac{FL^3}{48EI} \Rightarrow I_{min} = \frac{F \cdot (2280 \cdot 10^{-3})^3}{48 \cdot 210 \cdot 10^9 \cdot 0{,}01} = 0{,}00\,000\,2047\ m^4 = 2\,047\,000\ mm^4$$

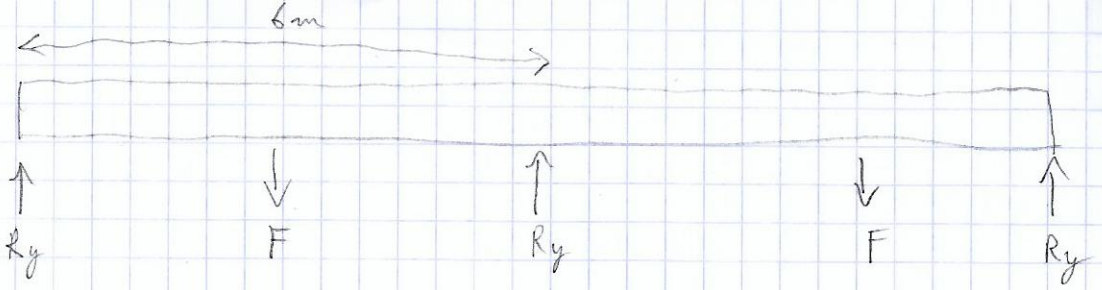
$1\,m^4 = (1000)^4\ mm^4$

$\Rightarrow$ HEA 100 $3{,}432 \cdot 10^6$

(HEA 120) $6{,}062 \cdot 10^6$

(3mm : 6822000)

6,822



$$F = (m_{\text{dwarsbalk}} + (2 \cdot m_{\text{vierkant}} + 8 \cdot m_{\text{hoekprofiel}})/2 + 2 \cdot m_{\text{rechthoek}} + m_{\text{AGV}})/2 \cdot g$$

$$= g(m_{\text{dwarsbalk}} + m_{\text{vierkant}} + 4 \cdot m_{\text{hoekprofiel}} + 2 \cdot m_{\text{rechthoek}} + m_{\text{AGV}})/2$$

aangepast:

$$= \frac{g}{2} \cdot \Big[7850 \big(\overbrace{0{,}15 \cdot 0{,}1722865 \cdot 0{,}813}^{17458935\ mm^3} + 0{,}4 \cdot 0{,}4 \cdot 0{,}015 + 4 \cdot \overbrace{0{,}000665}^{} + 2 \cdot 0{,}3 \cdot 0{,}15 \cdot 0{,}015\big) + 4 \cdot \underset{\substack{\cancel{0{,}73kg} \\ 5{,}45}}{m_{\text{lagen}}} + 4 \cdot \underset{\substack{7850 \cdot \underbrace{0{,}18 \cdot 0{,}78 \cdot 0{,}03}_{5238000\ mm^3}}}{m_{\text{rijplaat}}} + \underset{\substack{7850 \cdot 1 \cdot 1 \cdot 0{,}05}}{m_{\text{bodemplaat}}}\Big]$$

$$F = \frac{g}{2}[215{,}25 + 551{,}6768] = 3767{,}77\ N$$

$$\Rightarrow \overbrace{\sigma_{max}}^{\eta_{max}} \cdot W = \frac{F \overset{6m}{L}}{4} = \frac{F}{2} \cdot \frac{L}{2}$$

$N \cdot m \cdot \frac{m^2}{N}$

$$\Rightarrow W_{min} = \frac{F \cdot 6}{4} \cdot \frac{1}{\sigma_{toel}} = \frac{F \cdot 6}{4} \cdot \frac{1{,}5}{450\,MPa} = 0{,}0000188 09\ m^3 = 18{,}809\ cm^3$$

$\Rightarrow$ alle opties mogelijk

$$\Rightarrow \delta = \frac{FL^3}{48EI} \qquad \delta = (-1cm)$$

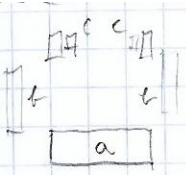
100^4

$1 \cdot (10^2)^4$

$$I_{min} = \frac{F \overset{6}{L}{}^3}{48 \underset{\substack{210GPa}}{E} \delta} = \frac{F \cdot 6^3}{48 \cdot 210 \cdot 10^9 \cdot 0{,}01} = 0{,}00000806 1\ m^4 = 806{,}1\ cm^4$$

$\Rightarrow$ PR 3 : 786,20 cm^4 $\Rightarrow \delta = 0{,}01012$

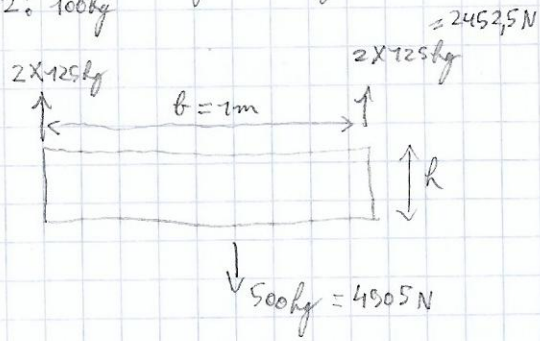
$\Rightarrow$ extra veiligheid $\boxed{\text{PR 4}}$ $\rightarrow$ 1370 cm^4

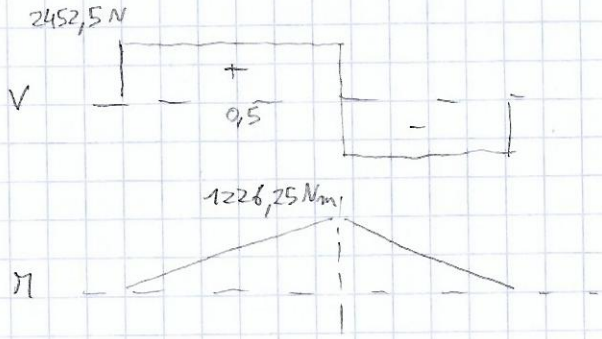


a) aanname: gewicht grootste last

M1: 350 kg + 150 kg (motor etc) = 500 kg

M2: 100 kg





1m x 1m x h

$W_x = \frac{b \cdot h^2}{6}$

$\sigma_{max} = \frac{M}{W_x} = \frac{6}{1 \cdot h^2} \cdot 1226{,}25\ Nm$

S235

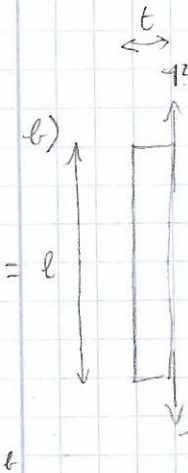
$\sigma_{toel} = \frac{\sigma_y}{\text{safety factor}} = \frac{235\ MPa}{1{,}5}$

$h = \sqrt{6 \cdot 1226{,}25\ Nm \cdot \frac{1{,}5}{235\ MPa}} = 0{,}00685\ m = 6{,}85\ mm$

$I = \frac{b \cdot h^3}{12}$

$\delta = \frac{FL^3}{48EI} = \frac{FL^3 \cdot 12}{48 E b h^3} \Rightarrow 0{,}01m = \frac{4905N \cdot (1m)^3 \cdot 12}{48 \cdot 210\ GPa \cdot 1m \cdot h^3}$ ($\delta = 1cm$ max) $\Rightarrow h_{min} = 0{,}00836\ m = 8{,}36\ mm$

→ h gekozen: 100 mm
voor bevestigingsruimte bouten



78,5 = 1m x 1m x 0,01m x ρ_{steel} = 7850 kg/m³

$(500\ kg + m_a)/4 = 578{,}5/4 = 144{,}625\ kg = 1418{,}77\ N$

$F \cdot \frac{t}{2}$ = moment ondat kracht niet centraal

$\sigma_{max} = \frac{N}{A} + \frac{M}{W_x} = \frac{1418{,}77\ N}{t \cdot 0{,}78m} + \frac{1418{,}77\ N \cdot \frac{t}{2}}{\frac{l t^2}{6}}$

$\sigma_{toel} = \frac{235\ MPa}{1{,}5}$

$\frac{1}{t}\left(\frac{F}{l} + \frac{3F}{l}\right) = \frac{1}{t} \cdot \frac{4F}{l}$

$\Rightarrow t = \frac{4F}{l} \cdot \frac{1{,}5}{235\ MPa} = 0{,}000201\ m = 0{,}201\ mm$

t_NX = 30 mm